

Fairpoint Regional Utility System Repairs 2,800 LF Failed 30-inch Water Main by Slipling with Fusible PVC® Pipe

Fairpoint Regional Utility System is a regional water supply entity located east of Pensacola, Florida in southern Santa Rosa County. The utility supplies potable water to several local water supply systems, including the City of Gulf Breeze, Holley-Navarre, Navarre Beach, and Midway. Fairpoint Regional Utility's system consists of seven aquifer supply wells located east of Milton and approximately 23 miles of water transmission mains conveying treated, potable water to the various coastal service areas. The transmission system was installed and placed into service in 2004.

A key segment of the transmission main is a 2,800 foot crossing under the East Bay. This section of the line was originally installed via horizontal directional drilling (HDD) using 30-inch DR9 DIPS HDPE pipe (32-inch OD, 24.5-inch ID). In the spring of 2014 the HDPE crossing was determined to be leaking and was taken out of service for inspection. A video of the interior of the pipeline revealed a gradual, spiraling, longitudinal crack 13 feet in length. Based on the original bore profile, the location of the crack was determined to start 408 feet from the south side of the bore at a depth of approximately 55 feet below the bottom of East Bay.



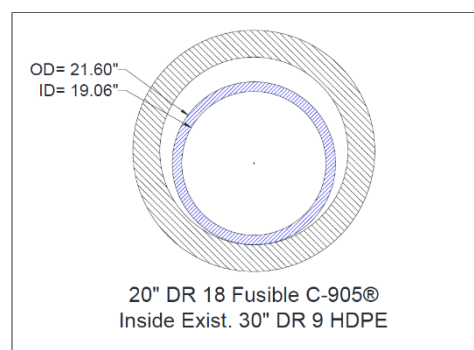
Photo of Longitudinal Crack

Ric Delp, Project Manager with Baskerville-Donovan, Inc., commented: "The cause of the failure is currently unknown. However, this is the **second reported failure on an HDPE bore crossing in the system.**"

In May 2014, an RFP was issued for the repair of the main. Two repair options were solicited, including: 1) a point repair of the failure area, and 2) a complete slipline of the 2,800 foot crossing using a new pipe. Two proposals were received, with both contractors proposing to slipline the entire bore section with a new, structurally independent pipe. Fairpoint reviewed the proposals and selected T.B. Landmark to slipline the failed line with 20-inch DR18 DIPS Fusible C-905® PVC pipe.

Pipeline Details and Project Summary

Project:	Fairpoint Regional Utility System Slipline
Location:	Navarre, Florida
Length:	2,800 LF
Pipe Size	20" DR18 DIPS Fusible C-905®
Installation	Slipline
Pressure Test:	150 psi
Engineers:	Baskerville Donovan, Inc., Municipal Engineering Services
Contractors	T.B. Landmark, Utility Services Company

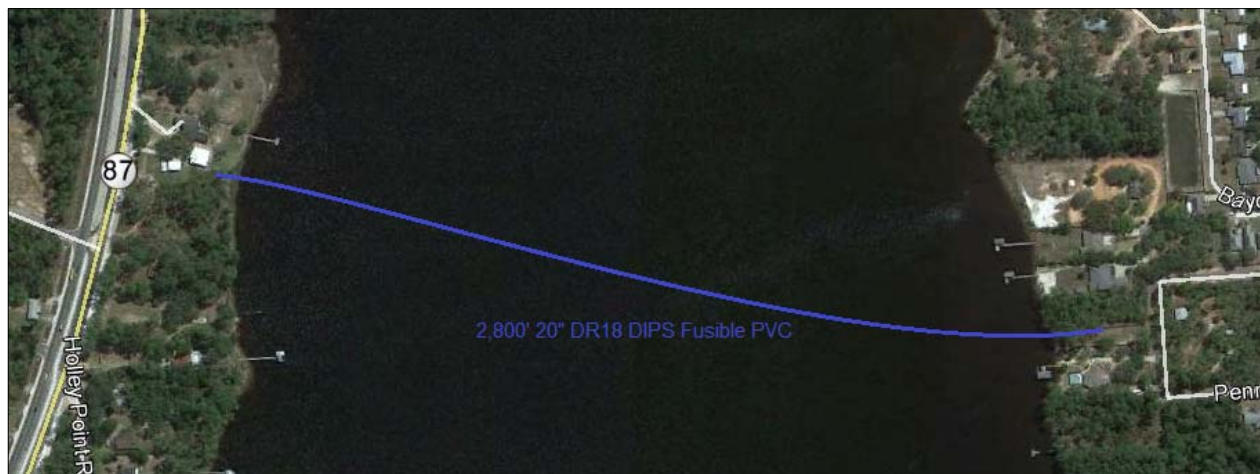


Cross Section of HDPE Host and Fusible PVC® Pipe

Underground Solutions, Inc. (UGSI) provides infrastructure technologies for water, wastewater and power cable conduit applications. UGSI's **Fusible PVC®** pipe products, including **Fusible C-900®**, **Fusible C-905®** and **FPVC®**, utilize patented technology to produce a fused monolithic, fully-restrained, gasket-free, leak-free piping system ideal for trenchless (horizontal directional drilling, pipe bursting and slipling) or conventional "open-cut" installations and are available in 4-inch to 36-inch diameters. The combination of standard fittings and lower weight with higher flow for a given pressure class versus other thermoplastic pipes ensures that Fusible PVC® pipe brings greater economy to most pipeline projects.

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T.B. Landmark's project plan called for mobilization of a horizontal directional drill rig to push a 22-inch rod ball through the existing 30-inch line to proof the pipe. The drill rig was set up on the north end of the bore alignment. Proofing of the existing pipe went smoothly and was completed in a single day.



An internal pull head was used to connect the drill stem to the Fusible PVC® pipe string. Due to space limitations on the south side of the bore alignment, the pipe string was fused and pulled into the host pipe in 45-foot sections in a “fuse-and-pull” installation.

Pull forces during installation never exceeded 17,000 lbs. The installation of the 2,800 foot pipe string required seven days to complete. According to Marty Adams of T.B. Landmark, “The pull forces required to install the pipe were significantly less than we anticipated.”

After completion of pipe installation, the new line was flushed and pressure tested at 150 psi for two hours. Following the pressure test, the line was disinfected and tested. Utility Services Company of Gulf Breeze, Florida made the final connections and tie-ins to place the transmission main back into service.

Phil Phillips of Municipal Engineering Services, Inc., Fairpoint's outside engineer, commented, “The ability to slipline this existing sub-aqueous bore with Fusible PVC® pipe enabled Fairpoint Utility System to rehabilitate the asset while maximizing the transmission main's hydraulic capacity.”



20" Fusible PVC® Pipe Entering
30" Host Pipe



Fused Sections of New 20" Pipe



Pull Head and Fusible PVC® Pipe
Exiting Host Pipe

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